

Understanding the multidimensional effects of product variety on customer behaviour: A large-scale empirical study in online fast fashion retail

Abstract: In collaboration with one of Europe’s largest fast fashion retailers, we study the implications of assortment variety on customer choice. We analyze a large, event-based clickstream dataset to characterize and quantify the effects of assortment variety on customer choice. We propose a novel definition and representation of assortment variety as a bipartite graph, which allows us to define variety along three dimensions: the number of styles, the number of colours, and density of the graph. We develop a *Random Consideration Set* (RCS) model, directly incorporating heterogeneity, in which we specify a nonlinear functional form for all variety variables, and parametrized as a Latent-Class Multinomial Logit (LC-MNL) model. We find our dataset has three main segments, with contrasting utility for economic variables, variety, and propensity to convert to a purchase. Our estimation results also show a combination of linear and nonlinear relationship between variety and customer utility. Our study further shows the importance of accounting for heterogeneity and deaggregation of variety into its multidimensional components in the modelling approach. Our findings highlight the inherent tensions of offering and efficiently managing large product variety to please different customer types, in addition to emphasize the trade-offs retailers must manage.

Key words: product variety, multidimensional variety, behavioural decision making, latent class model, clickstream data, fashion retailing

1 Introduction

The market size of the global fast fashion industry exceeded \$90 billion in 2022, and is expected to reach \$133 billion by 2026, representing a compound annual growth rate of 7.7% (Statista 2022). The success of the fast fashion business model lies in its ability to respond quickly to customers’ variety-seeking preferences. This ability results in quick turnarounds of assortments in fast fashion companies (Closa 2015). For instance, Zara releases new products every two weeks, for a total of approximately 10,000 new products a year, while SHEIN—which revolutionized *ultra*-fast fashion—releases as many as 6,000 new items daily (Segran 2021). Specific to the online channel, the offered assortment variety has exploded due to nonexistent physical display constraints and low inventory costs.

Product variety, however, poses operational challenges as it may lead to a small fraction of products collecting most of the views (and purchases), and the resulting heavy-tailed distribution (Brynjolfsson et al. 2011) makes forecasting difficult. Resultantly, product inventory is not always depleted, which leaves retailers with a large amount of unsold merchandise. For example, H&M reported more than \$4 billion worth of

unsold inventory in just one year, representing a 7% increase from the previous year (Paton 2018). These amounts exacerbate an already alarming waste crisis, which has tarnished the industry's public image and heightened the regulatory pressure on fashion brands. Overproduction and left-over inventory waste have been recognized as some of the biggest challenges for the fashion industry (Fashion Revolution 2023). While selecting an optimal assortment of products remains a hard task for any brand or retailer, a better understanding of consumer behaviour, in particular in regard to variety, is crucial to move the industry towards a sustainable future.

Additionally, there is growing evidence in the consumer psychology and marketing literature that, from a customer's perspective, more variety does not always lead to a positive experience and a purchase decision. In other words, variety may cause *choice overload* (Iyengar and Lepper 2000). Instead of helping customers find their preferred product, variety confuses customers and can make them abandon their shopping session. Moreover, variety in the fast fashion industry involves multiple dimensions (e.g., styles and colours). It is therefore crucial for brands and retailers to understand how these multiple dimensions can affect customer choice, and to balance the trade-offs when defining their product assortments.

The key question, then, is: *why is fast fashion accelerating?* The answer seems to be that the belief among retailers that customers prefer more variety persists. Two arguments support this belief. First, variety allows retail companies to address heterogeneity in taste across the market. In other words, a varied assortment increases the likelihood that customers will find a product to their liking. A large assortment is also stimulating, and provides customers with full information and flexibility for uncertain customers (Broniarczyk 2008). Second, assortment variety serves to hedge against the uncertainty in fast-changing fashion trends. Production decisions are usually made in advance of selling seasons and so more variety improves the odds of assortments matching the latest fashion trends (Long and Nasiry 2022).

In this context, we pose two main research questions. First, can we develop measures of product variety that capture its multidimensional nature, and that are readily and easily implementable on large and complex fashion assortments? Second, how are consumers affected by the multiple dimensions of this growing variety, and how does it vary across customer types? Can we find further empirical evidence to support the claim that more variety is not always better, and therefore help reduce the size of assortments to levels that are more sustainable?

To address these questions, we partnered with one of the largest fast fashion retailers in Europe that offers more than 30,000 unique style and colour combinations in every selling season. Our partner retailer's online channel provides us with a unique, large event-based clickstream dataset that captures snapshots of more than 400 million events (e.g., product view, add-to-basket, and checkout) generated by online customers visiting the retailer's online store. The dataset also contains firm-side operational data such as daily prices, discounts, and inventory levels. We use this rich and customer-specific dataset to micro-model the behaviour

of online customers—in particular, how they respond to the variety in the proposed product assortment—while controlling for (a) online price and promotion information posted on the page and (b) stock availability information.

We propose to represent assortment variety as a bipartite graph. In addition to overcoming the many limitations of similarity measures and product spaces, this representation allows us to define variety along three dimensions: the number of styles, the number of colours, and the density of the graph. These measures of variety are especially suitable for large and complex assortments, and are applicable for research in operations management, marketing, and consumer psychology research. In practice, our operationalization of variety has the advantage of corresponding exactly to the assortment decisions managers make and can be readily adopted.

We characterize and quantify the effects of product variety on customer utility by developing a Random Consideration Set (RCS) model which we estimate as a Latent-Class Multinomial Logit (LC-MNL) model. Our model defines the customer utility as a function of the information available on the category and product pages, of variety-specific contextual variables related to the assortment's bipartite graph representation, and control for potential endogeneity issues using the control function of Petrin and Train (2010) to isolate the effects of discount and availability (inventory levels) on customer choices. It further allows us to simultaneously study the impact of variety on the two stages of customer decision: the consideration and the choice stages.

We test our model empirically on customer-level clickstream data. We show the value of de-aggregating variety into its multidimensional components and in particular, we find contrasting preferences for styles, colours and normalized density between customer types and decision stages. We find that our dataset has three main segments, and that different segments display contrasting preferences for variety, but also for economic variables such as price and discount. In addition, we find that the relationship between variety and customer utility can be both linear and nonlinear, and when nonlinear, can be either U or inverted-U shaped. While we further provide empirical support that using an aggregate variety measure (i.e., total number of products) will result in observing an inverted-U shape relationship between variety and customer utility, our empirical results also show that de-aggregating variety into its multidimensional components significantly changes the empirical behavioural insights and further supports the idea that variety is a multidimensional construct. We also find that customers in our dataset have a strong preference for the colour black. We show that the likelihood of a style connecting to the colour black decreases as the number of styles increase, and that consequently, customer utility (and choice probability) significantly decrease. This paper consequently provides several managerial implications for fast-fashion retailers seeking to optimize their product variety along multiple dimensions. Our key results highlight the importance of adequately defining product assortment variety to capture the complex relationship to heterogeneous customer preferences. Ignoring to do so could lead managers to miss important nuances in customer preferences. Furthermore, our empirical results

on variety measures as defined in this paper can be efficiently used as guiding upper bounds by retailer managers when building assortments. With a better understanding of purchase behavior, we contribute to the greater goal of increasing product variety efficiency to achieve more sustainable resource consumption levels.

The rest of our paper proceeds as follows. In Section 2, we review the related literature. Section 3 motivates the business context and presents the measures of variety we use in our analysis. In this section, we also develop the theoretical model of customer behaviour. In Section 4, we describe the data in detail, test our model empirically, and discuss the implications of our results and segmentation. We conclude in Section 5 with a brief summary of our context and findings, as well as suggestions for future research.

2 Literature Review

In this section, we review the literature most closely related to our work.

Effects of Variety. The theory of rational choice assumes the relative valuation between two alternatives does not depend on the presence (or absence) of other alternatives—referred to as the independence of irrelevant alternatives (IIA) assumption (Luce 1959)—and implies that a consumer chooses the alternative with the highest utility from an offer set. An immediate implication of the IIA assumption is the regularity condition: the choice probability of all alternatives in the offer set decreases by the same relative amount as more alternatives are added to the offer set. When an offer set contains the choice of not purchasing a product (i.e., the no-purchase alternative), then adding more products to the offer set effectively decreases the choice probability of the no-purchase alternative. In other words, the theory of rational choice, through the regularity condition, suggests that more variety *increases* the likelihood of the assortment containing an option that matches a customer’s preferences and *decreases* the choice probability of the no-purchase option (Lancaster 1990). This intuition guided to the development of the fast fashion business model. It also increases the likelihood of product fit when customers have heterogeneous tastes. Research in operations management supports this idea. For example, Borle et al. (2005) show that a reduction in the number of stock keeping units (SKUs) leads to an overall decrease in retail sales. Ton and Raman (2010) and Kok and Şimşek (2021) show that increases in product variety and inventory levels result in higher sales. Literature on consumer psychology also notes many benefits to product variety. Variety is stimulating, especially for customers that exhibit a variety-seeking behaviour (Simonson 1990, Kahn 1995, Kahn 1998), and derive utility from having the ability to choose among multiple products in an offer set (Moe 2003). It further provides flexibility and opportunity to learn an offering for customers with uncertain preferences about a product category (Kahn and Lehmann 1991).

However, there is growing evidence in the marketing and consumer psychology literature that choosing from large assortments is cognitively taxing for consumers, hence why more variety does not always translate into either an enhanced experience or and increased likelihood of making a purchase. Iyengar and

Lepper (2000) show that customers are more likely to choose from a limited offer set rather than from an extensive one. Their study lead to the development of the *choice overload* concept (for a review, see Chernev et al. 2015). Several studies support this concept (see e.g., Chernev 2003b, Chernev 2003a, Gourville and Soman 2005). Broniarczyk et al. (1998) and Boatwright and Nunes (2001) argue that substantive reductions in the number of SKUs at a grocery retailer can be made without negatively affecting store choice (i.e., patronage) or category sales.

Past literature has only considered a linear relationship to develop the aforementioned insights, and while it has hinted at a possible nonlinear relationship between variety and consumer choice (Broniarczyk 2008), only recently models have included nonlinear terms (Wan et al. 2012, Wan et al. 2014, and Lu et al. 2022) and obtain an inverted U-shaped relationship between variety and sales, supporting the intuition that some level of variety is desirable and increases sales up to a threshold after which additional variety leads to lower sales. Our study specifies the customer utility as a quadratic function of variety-specific variables related to our proposed bipartite graph representation, which allows us to characterize—more accurately than before—the relationship between variety and consumer choice.

Research in consumer psychology has also established different effects of product variety on customer behaviour depending on which stage of the decision. Iyengar et al. (2006) and Chernev (2006) show that large product assortments seem to attract customers in the first stage, but hinder the second stage choice. Our empirical estimation using a LC-MNL model allows us to study this conflicting effect of product variety simultaneously.

Dimensions of Variety. Most commonly, variety is defined as the total number of products (or SKUs) in an assortment of a product category. While adopting the latter definition, Boatwright and Nunes (2001) propose an attribute-based model and show that SKU reductions were most effective when removing SKUs with duplicate attributes; doing so does not significantly reduce the assortment variety (in terms of available attribute levels) but rather declutters it. Similar results are also reported by Broniarczyk et al. (1998), who suggest that the perception of assortment variety is a multidimensional construct. van Herpen and Pieters (2002) also report that assortment size (in terms of total number of products) is not always a good proxy for perceived variety.

The preceding examples highlight the importance of defining assortment variety in a way that accurately reflects the concept's complexity. Studies that map product attributes to a product space (see e.g., Hoch et al. 1999, van Herpen and Pieters 2002, Gaur and Honhon 2006, Roederkerk et al. 2011) assume that the distance between products can be computed, therefore providing an easy measure of similarity between products. Despite being desirable and useful, this approach is often impossible. In many industries, such as fashion, the many product attributes are defined by unordered and not equidistant categorical variables (e.g., colour, bill of materials), and in such cases, it is impossible to define a product space. Roederkerk et al. (2011) discuss this issue as a limitation of their work. Our proposed representation of an assortment

as a bipartite graph allows us to conceptualize variety along three dimensions (or attributes): the number of styles, the number of colours, and the density of the graph. Additionally, it is suitable for applications in many fields, and can be used readily by managers.

Context Effects and Choices in Large Assortments. There is empirical evidence that consumers are not rational (Huber et al. 1982, Tversky and Kahneman 1991), but rather influenced by the context of the offer set in which the decision is made. In such context-dependent situations, the assumptions of the theory of rational choice (IIA and regularity) are violated, and substitution patterns can be significantly different. In a fast fashion retail setting, this can imply the choice probability of the no-purchase alternative may not be strictly decreasing as more products are added to the assortment. This is supported empirically (Broniarczyk 2008), showing that customers default the no-choice alternative when feeling overwhelmed by large assortments. Larger assortments also have as a consequence to dilute the product offering. As more products are introduced, the likelihood of introducing a new attribute or attribute level decreases. The result is a large assortment of weakly differentiated products, which increases confusion for customers, in addition to reduce the product fit past a certain threshold (Lehmann 1998).

Our proposed bipartite graph representation allows contextualizing variety-specific variables, and enable a richer and more accurate characterization and quantification of the relationship between variety and consumer choice. In the past, “large” assortments corresponded to assortments of a few tens of products (Broniarczyk 2008). Our study also provides an empirical context in which variety corresponds to $\sim 1,000$ products, and allows us to test if the insights previously found extend to the explosion in variety we have seen in the past decades.

3 The Empirical Model

In this section, we first discuss the business context that motivates our study. We then present our modelling framework and develop the measures to capture different dimensions of product variety.

3.1 Business Context

Our partner is a European fast fashion retailer operating brick-and-mortar stores and an online store. In this study, we focus solely on the online store operations.

The landing page of the online store displays different product categories for men, women, children, babies, and accessories. The website has a search function in addition to multiple banners promoting a range of products including new arrivals, sales, and seasonal essentials. A customer’s search eventually lands on a product category page (e.g., women’s dresses, sunglasses, sales) that displays, in a continuous scroll, thumbnail photos of products from the selected category along with product titles, available colours, sizes, and pricing information. Each product category page allows the customer to sort products. The default order is a combination of new arrivals and discounted products; a customer however has the choice to sort

by ascending or descending prices, and new arrivals among others. A customer can also choose to filter products on the category page by style, colour, size, and price range, or any combination thereof.

While scrolling through the chosen product category, a customer can access detailed information for a product by clicking on its thumbnail photo. Doing so takes the customer to the product page, which contains more pictures of the product, available colours, sizes and sizing information, product description, and information on shipping and returns. The customer can see the maximum number (usually 10) of units of a product she can order, but cannot see the product's inventory (i.e., how many units are in stock).

Production decisions are made well in advance of the selling season. Our partner retailer explained to us that they send one production request to their suppliers 4-5 months before the beginning of the selling season. This means that assortment variety and production volumes are set prior to realizing uncertainty in customer tastes during a selling season. Shipping decisions (from the suppliers to the retailers' main warehouses) are made solely on operational constraints (i.e., bundle sizing, shipping costs, etc.).

3.2 A Multidimensional Representation of Variety

We now present our proposed representation of product variety that explicitly incorporates its multiple dimensions. We define a product by the combination of a style and a colour. This definition is standard in the industry and followed by our partner retailer; it has also been used in the literature (e.g., Boada-Collado and Martínez-De-Albéniz 2020). We can then represent an assortment of products as a bipartite graph (see Figure 1). Such a graph has a set of style vertices and a set of colour vertices and each edge connecting a style vertex to a colour vertex corresponds to a product. One could employ similar representations in industries where the product variety is defined naturally along multiple dimensions, such as automobiles (models, colours, trim-levels) and food (flavor, processed vs. non-processed). The representation could also be extended to additional attributes (e.g., bill of materials), but we focus here on two dimensions for the exposition of the representation and variety measures.

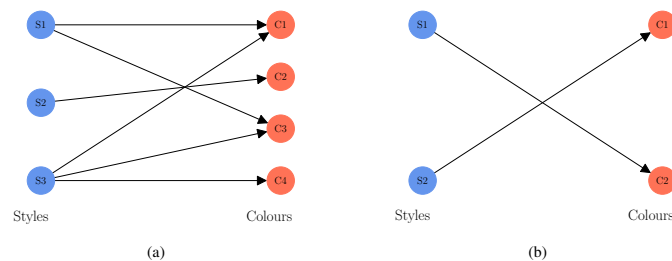


Figure 1 Bipartite Graph Representation of an Assortment

A key differentiating metric in our proposed graph representation is the density. One could see density simply as the total number of styles, and therefore think it is similar to the total number of products, a

variety metric that has been widely used in the literature (see §2). However, the density, and in particular the normalized density, allows a retailer to understand the level of differentiation in the assortment. Are all styles in all colours (i.e., maximal normalized density), or are style-colour combinations unique (i.e., minimal normalized density) and therefore more differentiated? Using the normalized density can help managers better understand how to combine different product attributes such that the offering is more appealing to customer. We define the normalized density of an assortment as:

$$\text{norm.density} := \frac{\text{density} - \text{density}_{\min}}{\text{density}_{\max} - \text{density}_{\min}} = \frac{\text{density} - \text{density}_{\min}}{1 - \text{density}_{\min}} \quad (1)$$

where $\text{density} = \frac{|E|}{|V_S| \cdot |V_C|}$, $\text{density}_{\min} = \frac{\max\{|V_S|, |V_C|\}}{|V_S| \cdot |V_C|}$, V_S and V_C are the set of style and colour vertices, respectively, $|V_S|$ and $|V_C|$ are the cardinality of these sets, and E is the set of edges in the graph with $|E|$ denoting its cardinality. As mentioned previously, the normalized density captures the added connectivity in a graph beyond the minimum density, which is itself a function of the minimum number of products in the assortment ($\max\{|V_S|, |V_C|\}$). Normalized density is bounded below by 0 if the graph is minimally connected (e.g., Panel (b) of Figure 1), and is bounded above by 1, when the graph is fully connected. In the case of a division by 0 ($\text{density}_{\max} = \text{density}_{\min}$), we assume the normalized density is equal to 0. Furthermore, the normalized density has the advantage to allow comparison across assortments that differ in order (number of vertices) and size (number of edges). To illustrate, consider the two assortments in Figure 1: both assortments have a density of 0.5. However, the graph in Panel (a) has a minimum density of 0.33, while the graph in Panel (b) has a minimum density of 0.5. A typical retailer must decide on the number of styles to design, the colour palette for the season, and which styles to offer in which colour. Note that the latter decision will impact how customers perceive variety, and can therefore significantly affect their choice behaviour. In our graph representation, these decisions correspond to choosing the vertices and how densely to connect the graph, and the normalized density measure provides the retailer with a gauge to how sparse (or dense) an assortment is. The bipartite graph representation has the additional advantage of offering a context-dependent definition of our variety measures, thereby indirectly capturing context effects.

3.3 Model Development

Classical random utility maximization (RUM) frameworks assume that decision makers (i.e., customers) have information about *all* alternatives before making a decision. In a market such as fast fashion, where product releases are frequent, and assortment sizes are substantial, this assumption simply does not hold. We confirm this from our data, as we see customers view on average 9 products (see Table 3), a number considerably smaller than the size of the dress assortment on any given date.

We adapt the *Random Consideration Set* (RCS) model proposed by Manzani and Mariotti (2014) to our clickstream data context in a way that is similar to that of Honhon et al. (2020) and Aouad et al. (2019). Our RCS model assumes customers have imperfect information about product fit, that is, they do not know

a priori which product from the offered assortment they prefer. The RCS assumes consideration sets for each decision maker to be random, that is, they are defined by a set of consideration probabilities over all alternatives in the offer set (assortment). This assumption neatly captures the inherent randomness in clickstream data and explains why many products will never be considered, and more importantly, that more attractive products may not be considered (because customers do not see them). In other words, our RCS model constitutes a model of random choices in which the probability of a decision maker choosing a given alternative is conditional on the decision maker being aware of the alternative in the first place, and of competing alternatives.

Consider a retailer that offers a set $S_t \subseteq N$ of products, indexed by $j = 1, \dots, n$, on date t . Initially, a customer lands on a product category page. In the consider stage, a random consideration set is generated. A customer may click on products in the offer set S_t , and we assume a customer adds a product to her consideration set if, from the category page, she clicks on the product to access the product page. A customer adds product j to her consideration set C_{it} with probability $\lambda_{it}(j|S_t, \text{variety}_{it}(S_t))$, where $\text{variety}_{it}(S_t)$ refers to the variety customer i experiences on date t . The vector λ_{it} thus represents click probabilities for all products in S_t for customer i (i.e., a distribution over alternatives). We understand the consideration probability as a combination of customer i being aware of alternative j and a limited set of competing alternatives (i.e., as being seen by customer i on the category page, which is a function of product placement, which is itself a function of variety), and of customer i being interested in the type of alternative j (as a function of individual characteristics). Consequently, we can express the probability of observing any possible consideration set as:

$$P(C_{it}|S_t, \text{variety}_{it}(S_t)) = \prod_{j \in C_{it}} \lambda_{it}(j) \prod_{k \notin C_{it}} (1 - \lambda_{it}(k)) \quad (2)$$

Similar to Moe (2006) and Honhon et al. (2020), we assume the population of customers is heterogeneous, and composed of M segments. We further assume that customers of a same segment will, due to having the same segment-specific utilities, have similar click propensities and alternative valuations. Consequently, we assume that $P(C_{it}) = P(i \text{ being in segment } m)$, and parametrize this probability as:

$$P(C_{it}) = P(i \text{ being in segment } m) = h_{it}(m|S_t, \text{variety}_{it}(S_t)) = \frac{\exp\{z_{it}\kappa^m\}}{\sum_{l=1}^M \exp\{z_{it}\kappa^l\}} \quad (3)$$

where z_{it} corresponds to customer-specific session and variety data (see §4 for a detailed presentation), and we set $\kappa^1 = \mathbf{0}$ for identification. A limitation to our data set is that we do not observe the product ordering, or the sorting and filtering options a customer might have applied on the category page. However, we see the full session clickstream behaviour and leverage this to define variables of z_{it} . Furthermore, similar to Honhon et al. (2020) and Aouad et al. (2019) we use the observed product page clicks as a proxy for realized consideration sets for each consumer in our data set.

In the choose stage, a customer chooses the alternative (including the outside option) in her (realized) consideration set that maximizes her utility. Similar to Aouad et al. (2019), we parametrize this choice probability as a Multinomial Logit (MNL) choice model on each customer's consideration set, for which we can define the probability of customer i choosing product j on date t , conditional on customer i being in segment m and her consideration set C_{it} as:

$$q_{it}(j|m, C_{it}, \text{variety}_{it}(S_t)) = \frac{\exp\{u_{ijt}^m(\text{variety}_{it}(S_t))\}}{\sum_{k \in C_{it}} \exp\{u_{ikt}^m(\text{variety}_{it}(S_t))\} + u_0} \quad (4)$$

where $q_{it}(j|m, C_{it}, \text{variety}_{it}(S_t)) = 0 \forall j \notin C_{it}$. We write the utility of product $j \in S_t$ for customer i on date t as:

$$U_{ijt} := x_{ijt} \beta^m + f(\text{variety}_{ijt}^k \delta^m) + \xi_{ijt}, \quad (5)$$

where β^m and δ^m are segment-specific coefficients, x_{ijt} are product- and customer-specific variables (see §4 for a detailed presentation of the variables), and $f(\cdot)$ is a function that reflects the effect of product variety on customer utility. The variety measures included in our specification relate to the bipartite graph representation of an assortment presented in §3.2 and include the number of style vertices tot.style_{ijt}^k , the number of colour vertices $\text{tot.colours}_{ijt}^k$, and the normalized density $\text{norm.density}_{ijt}^k$, where k denotes the subcategory to which product j belongs, as we assume the effect of variety on a customer choice within a product category is at the product subcategory level. We define a subcategory as a separation of products within a category under smaller subsets (e.g., within the dress category, there is a sleeveless dress subcategory, a strapless dress subcategory, a back-detailed dress subcategory, etc.). We assume the effects of variety on customer utility are nonlinear, and specify $f(\cdot)$ as a quadratic function as it provides us the flexibility of recovering a U- or inverted U-shape, or concave/convex increasing/decreasing relationship between utility and variety variables. We assume ξ_{ijt} are independent and identically distributed (*iid*) idiosyncratic Gumbell distributed shocks to utility. Furthermore, we define the utility of the no-purchase option as that $U_{i0t} = \beta_0^m + \xi_{i0t}$. Haaijer et al. (2001) show that not defining the utility of the “no-choice” alternative can lead to biased coefficient estimates.

Similar to Natan (2021), we assume that $|S_t| \perp \xi_{ijt} \forall j$, that is, the size of the assortment does not depend on consumers choices (sales) previously observed, but are rather pre-determined as a function of the production schedule (see §3.1). Similarly, we assume that $\text{variety}_{it} \perp \xi_{ijt} \forall j$. Furthermore, we assume there is no spillover information between alternatives, that is, customer i being informed about alternative j is independent of her being informed about any other alternative in the offer set S_t . This assumption is also used by Goeree (2008).

The unconditional probability of customer i choosing product j on date t given the offer set S_t can then be written as the expectation over segments:

$$\pi_{it}(j|S_t) = \sum_{m=1}^M h_{it}(m|S_t, \text{variety}_{it}(S_t)) \prod_{j \in C_{it}} q_{it}(j|m, C_{it}, \text{variety}_{it}(S_t))^{y_{ijt}^m} \quad (6)$$

Assuming each customer decision is independent of all others, the log-likelihood function defining the probability of each customer choosing the alternative observed in the data can be written as:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_t \sum_i \log(\pi_{it}(j|S_t)) = \sum_t \sum_i \log \left[\sum_{m=1}^M h_{it}(m) \left(\prod_{j \in C_{it}} q_{it}(j|m)^{y_{ijt}} \right) \right] \quad (7)$$

where $\boldsymbol{\theta}$ corresponds to the vector of coefficients we are trying to estimate, including both first- (κ^m) and second-stage ($\beta^m, \delta^m, \eta^m$) coefficients, and $y_{ijt} = 1$ if customer i , chooses alternative j on date t , 0 otherwise.

The parametrization of our clickstream and choice model in eq.(7) above can be recognized as the Latent Class MNL (LC-MNL). The LC-MNL has been used in several studies in OM and economics (e.g., Bhat 1997, Greene and Hensher 2003) to model heterogeneity in choice situations, and has the advantage of not suffering from the IIA. We use the EM algorithm to estimate model coefficients. We choose this algorithm over direct maximization of the log-likelihood function due to the ease of implementation given the size of our dataset and the computational stability of the estimation. The main steps of our EM algorithm are presented in Algorithm 1 (see Appendix EC.2). We obtain standard errors for all coefficients by directly evaluating the Information matrix of Step 2 and Step3 at MLE point estimates.

A Note on Fixed Effects. Contrary to classical applications of customer choice models (e.g., preference surveys, conjoint experiments estimated using an aggregate MNL model), the clickstream data (collected during the online store's natural course of operations) and the modelling approach we use to estimate model parameters do not possess a panel structure. Therefore, there is little to no repeated cross-section structure in the data, which explains why our model does not include product-level fixed effects. As highlighted by Brownstone et al. (2000) (who were estimating an aggregate MNL model), estimating choice models with large choice sets (in their study, they have 698 alternatives, half of what we have in our data) creates computational difficulties (due to ill-conditioned Hessian matrix), and solutions do not always lead to robust results. Furthermore, we do not include subcategory fixed effects since the variety variables included in our specification are at the subcategory level, and therefore any subcategory fixed effect would be perfectly correlated with those variables. Finally, we do not include time (date) fixed effects either because we are observing each customer exactly once and so there can be no unobserved customer-specific and time-specific effects in our data, in addition to the fact that variety variables are normalized to time by definition.

3.3.1 Model Selection.

We repeat the estimation procedure for a successively increasing number of segments and calculate the Bayesian Information Criterion (BIC) for each model as a measure of model fit. The optimal number of segments M corresponds to the model for which the measure of fit stops improving (a lower value of BIC indicates a better fit). We compute the BIC as $BIC = -\mathcal{L}(\boldsymbol{\theta}) + 0.5R\log(N)$ where the first term corresponds to the full log-likelihood of the model, R corresponds to the number of coefficients estimated (i.e., cardinality of $\boldsymbol{\theta}$), and N corresponds to the total number of observations (i.e., customers) in our dataset.

3.3.2 Endogeneity.

Under model specification (5), some variables may be correlated with the idiosyncratic shocks ξ_{ijt} . In particular, decisions made by the retailer during the selling season are subject to endogeneity. We believe that only two variables, *discount* and *broken assort*, are potentially endogenous. The base price of the product is a decision that is made prior to the selling season and therefore does not vary.¹ In addition, our partner retailer has confirmed that the release of new products is scheduled ahead of time, and therefore, variety measures are not dependent on sales.

Endogeneity in the *discount* and *broken assort* variables may lead to biased estimates of the model's coefficients. To address this potential endogeneity issue, we follow Petrin and Train (2010) and use the two-stage *control function* approach. In the first stage, we define two linear regression models, one for each potentially endogenous variable, and regress the endogenous (either *discount* or *broken assort*) variable on all model variables and an appropriate instrument w_{ijt} . That is:

$$\begin{aligned} discount_{jt} &:= u'_{ijt} + \gamma_3 broken_assort_{jt} + \mu^{discount} w_{ijt}^{discount} + \mu^{broken_assort} w_{ijt}^{broken_assort} + \zeta_{ijt}^{discount}, \\ broken_assort_{jt} &:= u'_{ijt} + \gamma_3 discount_{jt} + \mu^{discount} w_{ijt}^{discount} + \mu^{broken_assort} w_{ijt}^{broken_assort} + \zeta_{ijt}^{broken_assort}, \end{aligned}$$

where $u'_{ijt} := \gamma_1 price_j + \gamma_2 age_{jt} + f(variety_{ijt}^k)$. In the first-stage regression, we omit the variable *view.count* because it is a customer-specific variable. In the second stage, we include the first-stage residuals $\zeta_{ijt}^{discount}$ and $\zeta_{ijt}^{broken_assort}$ in specification (5). By adding the residuals of the first-stage regressions to the specification, ξ_{ijt} are now independent of all other covariates. Specification (5) now becomes:

$$U_{ijt} := x_{ijt} \beta^m + f(variety_{ijt}^k \delta^m) + \eta_1 \zeta_{ijt}^{discount} + \eta_2 \zeta_{ijt}^{broken_assort} + \xi_{ijt}. \quad (8)$$

The control function approach requires the covariates of specification (8) and instruments $w_{ijt}^{discount}$ and $w_{ijt}^{broken_assort}$ to be independent of the residuals $\zeta_{ijt}^{discount}$ and $\zeta_{ijt}^{broken_assort}$, but do not require that the covariates and the instrument be independent of each other. Lagged variables have been used as instrumental variables in similar settings (e.g., Tan and Netessine 2014, Chuang et al. 2016, Boada-Collado and Martínez-De-Albéniz 2020). We use the lagged discount rate $discount_{ij,t-1}$ as an instrument for *discount* because the discount rate of the previous date $t - 1$ does not directly affect the customer at time t , as the customer makes a choice based on the discount she sees on date t . We justify the use of the lagged broken assortment indicator $broken_assort_{ij,t-1}$ as an instrument for *broken assort* for the same reasons.

4 Empirical Analysis

In this section, we describe our data and examine the multidimensional effect of variety on customer utility.

¹Contrary to some studies (e.g. Berry et al. 1995), we understand from conversations with our partner retailer that the base price of products is not influenced by unobserved characteristics, and instead reflects cost and market competition.

4.1 Data

Our analysis leverages two unique datasets provided by our partner retailer: (i) a large event-based clickstream dataset containing customer visit sessions, and (ii) a limited firm-side dataset containing daily information on pricing, inventory, and product attributes.

We apply our model to data from the women's dress category. Approximately 60% of the products with which customers interact in our clickstream data are from the women category, and the dress category is the largest and most active category for women. This was confirmed in discussions with our partner retailer. We therefore select only customers who viewed dresses during their session (i.e., they had dress products in their consideration sets). For tractability, we restrict our study to the period from February 19, 2018, to May 27, 2018. This three-month period corresponds to the online store's regular selling season. Figure 2 illustrates the evolution of the daily total number of dresses available at the online store over those three months. The figure clearly shows a ramping up of product introduction between February 19 and April 8, followed by a plateau until a sharp decrease around the end of May when the liquidation season starts. The two blue dashed lines represent the beginning and end of the three-month period.

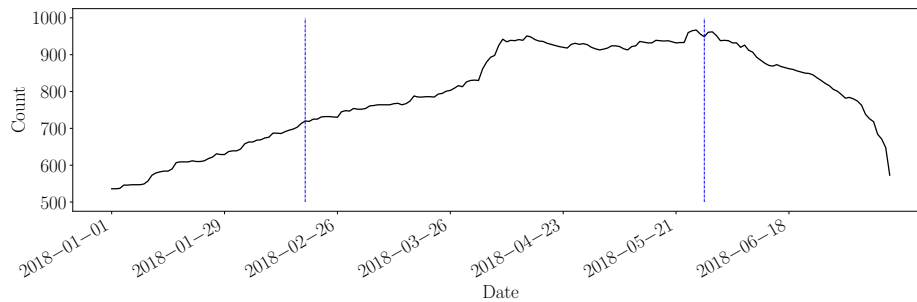


Figure 2 Evolution of Daily Number of Unique Dress Products Available

Clickstream Data. Clickstream data were collected directly from the retailer's website and includes snapshots of more than 400 million events such as page view, add to basket, checkout, etc., generated by visitors to the site. These data were collected between January 2017 and December 2018. The clickstream data describes customers' browsing, from the moment they arrive at the retailer's online store, to the moment they leave (whether they make a purchase). The granularity of the data allows us to see every step of a customer's journey through the website. In particular, the actions we extracted from the source files correspond to browsing clicks (home page, main page, category page, brand page), search clicks (search page), interaction clicks (page view, product view, basket add or remove), and checkout clicks (purchase event). For each click event, we collect following information: unique anonymized user identification number, time stamp, click type, and details about the click type (e.g., the name of the category page, promotion banner, unique product identification number), and product information. For basket events, such as add-to-basket

or checkout, the following additional information is collected: product list, unit price, quantity (units) in the basket, and total basket price.

The unique anonymized user identification is generated by cookies. One limitation of this data is that, if a user clears her cookies between visits, then the website can no longer recognize her, and she will be assigned a new unique identification number. We assume the frequency at which this happens is low.

We also note that the clickstream data were collected at the product level and does not contain any sizing information (an SKU is the combination of a product and a size), even for basket events. As such, it is not possible from our data to know which SKU a user purchased. This difference does not limit us in our application, however. We argue that size does not constitute an attribute on which a user will make a preference choice—as opposed to a style, colour or pattern, but is rather similar to the stock availability of an item. If a user finds an item she likes, but her size is not available, she will move on to other styles, or leave. This is naturally captured by the clickstream data. In addition, and most importantly, the focus of the study is on the effects of product variety on customer choice.

Firm Data. The second dataset consists of firm-side data and contains daily pricing and inventory information at the SKU level, but limited product information such as the subcategory the retailer assigned each product to, styles, and colours. We do not have the full list of attributes and levels that define each product, and therefore resort to defining a product as a combination of a style and colour.

Data Cleaning and Preparation. The clickstream data were collected in the natural course of operations of the online store. Unlike data collected in the process of a controlled experiment (e.g., Aouad et al. 2019), the clickstream data we obtained from our partner retailer is rich, yet noisy. Consequently, we devoted a considerable effort to extracting, cleaning, and preparing the clickstream data to retain its richness, while allowing a tractable implementation of the estimation procedure. The data cleaning and preparation, including handling outliers and imputation of firm data, is detailed in EC.1.²

We focus on a customer's first session at the online store to remove any complementarity or sequential effects that might affect a customer to return to the online store to purchase a product she saw previously. Alternatively, we are focusing on customers' first impression of an assortment, and how it affects their choice. The procedure results in a final dataset with 1.6 million observations from 164,000 unique customers who considered 1,209 unique products.

Segmentation and Choice Variables. Our estimation procedure requires two groups of variables: vector z_{it} of observed session click behaviour and variety measures (aggregated at the customer level) and vector x_{ijt} of observed product attributes and variety measures (at the customer-product level). We build all variables included in our model from clickstream data. Sample data, summary statistics and correlation matrices of each set of variables is presented in Appendix EC.1.3.

²All codes and notebooks used to generate data—redacted to preserve proprietary information—are available upon request. Sample data used for the choice model estimation algorithm is presented in EC.1.

The second stage variables and model are meant to capture the probability of a customer choosing a product within her consideration set. We include product-specific as well as product-and-customer-specific variables as they have been shown to be good predictors of choice. We include the base price (*price*) of products as a signal of overall product quality, the discount rate (*discount*) ranging between 0 (no discount) and 1 (fully discounted), and the age (*age*) of the product to capture for how many days it has been available. We further include a binary variable for broken assortments to control for the missing size information in our data and for the chance a customer finds her preferred product stocked out. We define *broken.assort* as a proxy variable to control for the possibility that a customer may find her preferred product stocked-out, leading her to not purchase the product (and either substitute for the next best product, or leave without a purchase). *broken.assort* is equal to 1 if at least one of the sizes is stocked out, and 0 otherwise. Finally, we include the number of times a customer viewed a product in her consideration set (i.e., has accessed the product page) as *view.count*. We include this variable as it has been observed that customers that return to a product more often are more likely to purchase it.

We define the relative variety as the variety (i.e., total number of styles, colours, or normalized density) within a subcategory normalized by the total variety (i.e., total number of styles, colours, or normalized density) in the dress category on that date. In other words, we view styles and colours as the two attributes that define a product, then the relative variety measures represent the proportion of levels of each attribute each subcategory contains (over the total number of levels in a category on a given date). Subcategories are used by the retailer for managerial purposes (e.g., different teams may be in charge of design, production, and logistics), but are not apparent to the customer. For example, there can be a sleeveless dress in the floral dress category, and a floral patterned dress in the sleeveless dress category. Furthermore, such a normalization allows us to compare variety across dates easily. Following this normalization, we rescale the variables such that they range between 0 (i.e., a subcategory which, on date t , has the least number of styles colours or normalized density) and 1 (i.e., a subcategory which, on date t , has the most number of styles colours or normalized density). Given that we observe some subcategories with only a handful of styles or colours on any given date (the minimum number of styles and colours within a subcategory is 1, see Table EC.2 and EC.5), a relative variety of 0 corresponds to no variety.

The first stage variables and model are meant to capture the probability that a product, out of all products on the category page, is being considered (i.e., clicked on). Recall the only information available to the customer at this stage when they land on the category page are thumbnail pictures of the products with pricing, discount, and colour information. Also recall that the default ordering (as well as other ordering options) are a combination of price (and discount) and age. We therefore include as what we label *product placement* control variables the median *price*, *discount*, and *age* a customer considers as an indication of she interacts with the product placement. The variety variables are introduced because products from larger subcategories have a higher probability of being seen by the customers, yet all products are competing

against each other. Note that the two sets of variety measures, one per stage of our model, allows us to simultaneously study the impact of variety on consideration and choice (conditional on consideration). We also add a segment constant to our first-stage model.

Finally, all variety variables in both stages are centered for numerical stability of the EM algorithm as including square terms results in high collinearity otherwise. The estimation results are re-centered and standard errors are calculated using the Delta method, resulting in re-scaling the linear terms only, the constant in the first stage, and the addition to a constant to the variety function $f(\cdot)$ of eq. (8).

4.2 Empirical Findings

We estimate 4 models ($M = 2, 3, 4, 5$) and find the best model fit to be for the 3-segment model. Table 1 compares the BIC for each model. Table 2 presents our main findings, by stage and segment.

Table 1 Model Fit Comparison

M	2	3	4	5
BIC	20,735.28	20,717.62	20,775.00	20,874.66

Table 2 Estimation Results

		Segment 1		Segment 2		Segment 3	
		Estimate	St.Err.	Estimate	St. Err.	Estimate	St. Err.
First Stage (consideration)	<i>med.cs.price</i>			-4.445E-03***	1.985E-04	4.630E-05	1.780E-04
	<i>med.cs.discount</i>			0.015***	4.489E-04	-1.025E-03**	4.319E-04
	<i>med.cs.age</i>			-8.737E-04***	2.180E-04	3.738E-04*	2.105E-04
	<i>med.cs.subcat.rel.styles</i>	Base segment		0.814***	0.185	0.146	0.175
	<i>med.cs.subcat.rel.styles²</i>			-0.394**	0.161	-0.192	0.156
	<i>med.cs.subcat.rel.colours</i>			-1.867***	0.176	-0.117	0.174
	<i>med.cs.subcat.rel.colours²</i>			2.571***	0.172	0.111	0.175
	<i>med.cs.subcat.rel.norm.dens</i>			-1.888***	0.368	0.195	0.347
	<i>med.cs.subcat.rel.norm.dens²</i>			3.424***	0.571	-0.651	0.609
	constant			0.164***	0.050	7.373E-03	0.047
	<i>price</i>	-0.014***	2.741E-03	-5.372E-03***	4.88E-04	-0.047***	4.742E-03
	<i>discount</i>	0.012**	5.584E-03	8.411E-03***	1.07E-03	0.039***	6.163E-03
	<i>age</i>	0.012***	5.584E-03	8.411E-03	1.07E-03	0.039***	6.163E-03
Second Stage (choice)	<i>borken.assort</i>	0.180	0.310	-0.330***	0.048	0.506**	0.240
	<i>view.count</i>	0.366***	0.037	1.010***	0.012	0.448***	0.041
	<i>rel.styles</i>	5.603**	2.422	0.950*	0.522	-61.162***	6.227
	<i>rel.styles²</i>	0.398	2.125	-0.175	0.418	46.027***	4.953
	<i>rel.colours</i>	1.841	3.280	-1.152**	0.518	93.386**	8.511
	<i>rel.colours²</i>	-11.335***	4.360	0.381	0.416	-57.834***	5.406
	<i>rel.norm.dens</i>	16.488***	4.509	-1.829**	0.816	8.481	8.317
	<i>rel.norm.dens²</i>	-37.838**	16.145	1.313	1.335	4.488	48.503
	<i>delta.const</i>	-1.134*	0.648	0.184*	0.103	-14.110***	1.431
	<i>resid.discount</i>	0.063***	8.642E-03	-7.723E-03***	4.325E-03	-0.058***	0.017
	<i>resid.ba</i>	-3.645***	0.344	0.598***	-4.445E-03	-0.283	0.518
	<i>no.choice</i>	8.766***	0.475	5.764***	0.085	14.620***	0.951
Segment size		5.55%		23.50%		70.95%	
Conversion rate		0.48%		8.23%		0.085%	
Observations		1,595,492					
Log-likelihood		-20,345.39					
McFadden R^2		0.943					

*: $p \leq 0.1$; **: $p \leq 0.05$; ***: $p \leq 0.01$

First-Stage Results. Our first-stage results show that almost all coefficients for Segment 3 are statistically not significant, suggesting there are no differences between Segment 1 and 3 in terms of customer

segmentation. On the other hand, all coefficients for Segment 2 are highly statistically significant, suggesting that customers from Segment 2 react differently to the variables included in our model than customers in Segment 1. We discuss the economical interpretation of our 3-segment model in §4.3.

For Segment 2, our results show that customer preferences for variety varies nonlinearly. Figure 3 illustrates the estimated relationship between variety variables and their contribution to segmentation, based on coefficient estimates, with 95% confidence interval bands (shaded area). Table 3 presents summary statistics of first-stage variables by segments. We also include in this table summary statistics for the consideration set as well as summary statistics for the median values of variables used in the second stage.

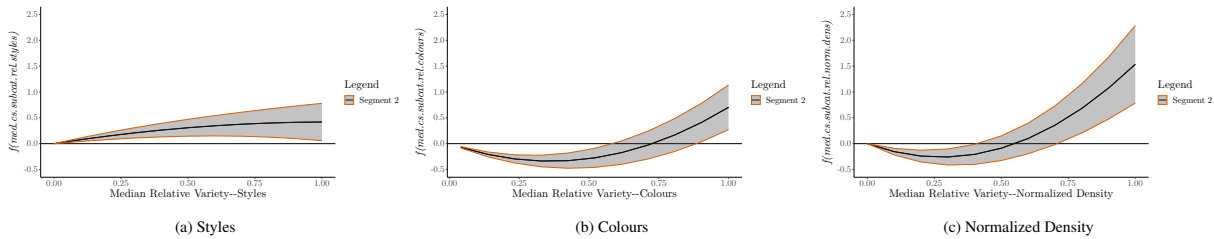


Figure 3 Estimated Effects of Variety on Customer Utility—First Stage

Table 3 Segmentation Variables—Summary Statistics

	Segment 1				Segment 2				Segment 3			
	Mean	St. Dev.	Min.	Max.	Mean	St. Dev.	Min.	Max.	Mean	St. Dev.	Min.	Max.
<i>consideration.set</i>	9.346	5.181	5	32	8.089	3.883	5	32	8.999	4.772	5	32
<i>med.cs.price</i>	118.26	32.81	24.99	249.99	107.34	25.03	24.99	274.99	134.86	41.60	24.99	299.99
<i>med.cs.discount</i>	0.30	0.17	0	0.65	0.39	0.14	0	0.71	0.17	0.19	0	0.64
<i>med.cs.age</i>	59.012	33.416	1	173	89.514	35.426	1	173	64.499	41.188	1	173
<i>med.cs.view.count</i>	1.11	0.347	1	7	1.05	0.212	1	6	1.06	0.243	1	9
<i>med.cs.broken.assort</i>	0.862	0.327	0	1	0.911	0.269	0	1	0.774	0.399	0	1
<i>med.cs.rel.styles</i>	0.303	0.258	0	0.978	0.394	0.241	0	1	0.265	0.094	0	0.948
<i>med.cs.rel.colours</i>	0.395	0.155	0.036	0.984	0.492	0.227	0.036	1	0.346	0.128	0	0.979
<i>med.cs.rel.norm.dens</i>	0.034	0.065	0	0.688	0.022	0.045	0	1	0.028	0.026	0	0.414

Panel (a) shows a concave increasing function between relative styles and customer segmentation. We interpret this relationship as the probability of being in Segment 2 increasing as the median relative styles a customer interacts with during her consideration stage. In other words, customers in Segment 2 prefer more variety in the first stage of their decision (than customers in Segment 1). We see that the increase in probability is diminishing as relative variety increases, and seem marginal beyond 0.75. Panel (b) and (c) both show U-shape relationships suggesting the probability of a customer being in Segment 2 (over being in Segment 1) as being lower than 1 for low levels of relative variety, but increases rapidly at medium and high levels of relative variety. We see that the probability becomes greater than 1 at 0.363 for relative colours, and 0.276 for relative normalized density. We interpret these results as customers in Segment 2 preferring high levels of relative colours during their consider stage, and summary statistics of Table 3 support this observation as we see customers in Segment 2 have the highest mean and maximum median

relative colours in their consideration sets. The relationship in Panel (c) for relative normalized density, however, differs in interpretation. While summary statistics of Table 3 show the average relative normalized density is smallest for Segment 2, they also show that only Segment 2 customers interact with the highest levels of relative normalized density. Consequently, Segment 2 customers, on average prefer lower levels of relative normalized density during the consideration stage (as indicated by the low average), but also contain all the customers that interact with the highest levels of that variety variable. We discuss these observations in more detail in §4.3.

Second-Stage Results. Our second-stage results show that most of the non-variety variables are highly statistically significant and that their signs are consistent with prior expectations. We discuss deviations from those expectations in §4.3. Recall that the variables *discount* and *broken assort* may be endogenous. Our results show that η_1 and η_2 , the respective coefficient estimate for *resid.discount* and *resid.ba* in eq. (8), are statistically significant, suggesting that both variables are endogeneous, and that we require including them to consistently estimate model coefficients. Our results show a mix of linear and nonlinear relationships between customer utility and variety variables. Figure 4 illustrates the estimated relationship between variety variables and their contribution to customer utility, based on coefficient estimates, with 95% confidence interval bands (shaded area). We do not include the relationship of Segment 3 and relative normalized density as the estimated coefficients are not statistically significant.

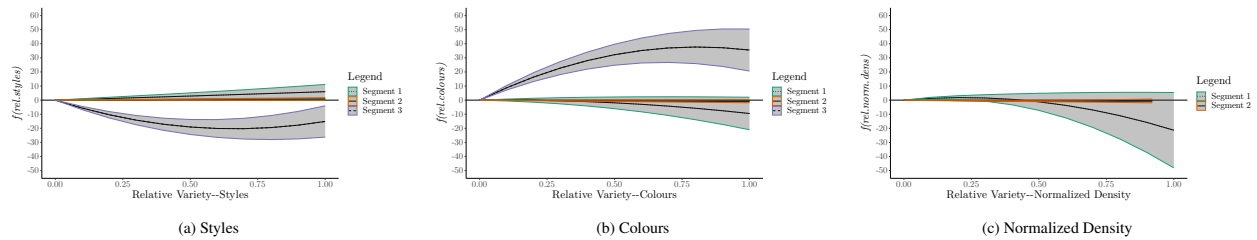


Figure 4 Estimated Effects of Variety on Customer Utility–Second Stage

Panel (a) shows that the relationships of Segments 1 and 2 towards relative styles is linearly positive, and that the effects of relative styles on Segment 2 are marginal, while they are slightly larger for Segment 1. Panel (a) further shows a U-shaped relationship for Segment 3, for which the effects of relative styles is largest among the three segments. The overlapping confidence interval bands between Segments 1 and 2 indicate there is no significant differences between the two segments. However, the non-overlapping confidence interval bands between Segments 1 and 2, and Segment 3 indicate a significant difference. The nonlinear relationship of Segment 3 suggests that customers in that segment prefer lower levels of relative styles and that their utility decreases up to a certain threshold, after which the utility starts increasing again. We discuss in more details the interpretation of the decreasing part of the relationship in §4.3. To provide a rationale for the increasing part, we look at the distribution of sales (in units) by the relative styles within

a subcategory in our data to provide a rationale for the increasing part of the nonlinear relationship for Segment 3. We test that this increase for high levels of styles could be attributed to demand creation. We find that all sales happen in subcategories with a relative style less than 0.664 (the minimum of the relationship) except for one sale that happens in a subcategory with a relative style of 0.969. We conclude that the increasing part we see in Panel (a) may be attributed to this “outlier” and the quadratic specification, rather than demand creation.

Panel (b) shows a nonlinearly decreasing relationship for Segment 1, a negative linear relationship for Segment 2, and an inverted U-shaped relationship for Segment 3. Again, we see that the effect of relative colours on customer utility is lowest for Segment 2, slightly larger for Segment 1, and largest for Segment 3. Furthermore, the overlapping confidence interval bands between Segments 1 and 2 indicate there is no significant differences between the two segments. However, the non-overlapping confidence interval bands between Segments 1 and 2, and Segment 3 indicate a significant difference. The negative relationships for Segments 1 and 2 suggest that, within a subcategory, a retailer is worse-off offering two colours rather than one, three rather than two, and so on. To provide a rationale for this observation, we look at the distribution of sales (in units) by colour in our data (see Table EC.11 in Appendix EC.5). We observe that 41% of the products purchased were in the colour black. The second most purchased colour is red with only 7.6% of all sales, suggesting customers shopping for dresses at our partner retailer have a strong preference for the colour black, which helps explain the result in Panel (b): as the number of colour vertices increases, the likelihood of finding the preferred style linked to the preferred colour (in our case, black) decreases. We verify this assumption by examining the relationship between the total number of colour vertices and the normalized vertex degree, which we define as the ratio of the degree of a colour vertex and the total number of styles in the subcategory (the total number of styles is the maximum possible degree of that colour vertex). Our correlation analysis shows that there exists a negative correlation ($r = -0.509$) between the total number of colour vertices and the normalized black colour vertex degree. That is, as the number of colours in the assortment increases, the fraction of products with black colour decreases (see Figure EC.1). The inverted U-shaped relationship for Segment 3 has its maximum a relative colour of 0.807, which translates to approximately 23 colours to choose from. Interestingly, this corresponds to almost twice the number of basic colours. Although it may be purely coincidental, the correlation is still worthy to note, and could suggest that at a higher level of relative colours, customers stop differentiating shades of colours, and rather lump them as one.

Finally, Panel (c) shows a slightly inverted U-shaped relationship for Segment 1, and a negative linear relationship for Segment 2. The non-overlapping confidence interval bands suggests there is a significant difference between Segment 1 and 2 from 0 to ~ 0.25 , but no significant difference for higher levels of relative normalized density due to the overlapping confidence interval bands. While the relationship for Segment 2 is almost flat, suggesting customers in this segment are close to indifferent to this dimension of

variety (like customers in Segment 3), the relationship for Segment 2 marginally increases above 0 between 0 and ~ 0.4 , with an optimum at 0.218, before decreasing rapidly. This latter observation suggests that customers in Segment 1 value low levels of relative normalized density (i.e., sparser assortments).

Comparisons with Alternative Model Specifications. We compare results of our 3-segment model with those of an aggregate MNL model (i.e., without any segmentation) on all customers' consideration sets, as well as with a LC-MNL model using as a variety measure the relative products (derived from the total products in a subcategory normalized by the total number of products in the dress category), referred to as an aggregate variety measure, corresponding to the most common modelling approach found in the literature. Full estimation results for both models are presented in Appendix EC.4.

Comparison of the results of our model to those of an aggregate MNL model in Table 4 shows that directly including heterogeneity via segmentation in the modelling approach improves the fit of the model, but more importantly, highlights how much behavioural insights can be missed by an aggregate MNL model, particularly in terms of the effects of variety variables on customer behaviour and the preference for the “no-choice” alternative. Results of the MNL model show that variety affects customer utility linearly, as we observe for Segments 1 and 2. However, the latter two segments account for only a quarter of the total customers, and are least impacted by variety (in the choice stage), which could result in a skewed perception of customer preferences when not incorporating heterogeneity in the modelling approach. Missing such insights could result in lower customer satisfaction and consequently, poorer business performance. Another important difference is the estimated preference for the “no-choice” option. Results from the MNL model show a preference close to that of Segment 2, which has the highest propensity to convert. However, the overall conversion rate in our dataset is 2%, largely below what is observed for Segment 2. This could lead to an erroneous expectation that customers may be more likely to convert to a purchase than they really are. Comparison of the results of our model to those of a LC-MNL using an aggregate measure of variety in Table 4 shows little differences in the overall fit of the model, but highlights important differences in the segmentation and the effect of variety on customer utility. We also find that a 3-segment model is optimal when using an aggregate variety measure, however, the segmentation is different, which leads to large difference in estimated utilities for variety variables, as well as conversion rates for each segment.

To summarize, our results provide evidence that different dimensions of variety affect customers differently, and that not accounting for heterogeneity on the customer population or only looking at variety in aggregate terms may lead researchers and managers to miss important nuances in customer preferences. It further highlights the need for adequate measures of assortment variety; measures that will capture its complex effects on customer utility. More importantly, our results show that the direction, shape and magnitude of the effects are vastly different from one dimension to another, from one segment to another, and from one decision stage to another. Our results also show that variety affects customers nonlinearly during their consider stage (for Segment 2), and show that customer preferences for variety vary linearly for the two

smaller out of three segments in the choice stage. Our results show that customer preferences for variety vary nonlinearly for the largest customer segment.

4.3 Economic Interpretation of the 3-Segment Model and Discussion

We provide in this section an interpretation of each segment in terms of customer behaviour, as well as a detailed discussion of the results of the previous subsection. Our discussion highlights the differences between segments and decision stages, and the impact of variety variables on each. Furthermore, it allows for a richer understanding of the tension that exists between large assortments, customer preferences, and business performance, in addition to put forward the trade-offs retailers face in choosing the optimal assortment to offer to customers.

Contrary to what has been previously thought (Broniarczyk 2008) and shown (Wan et al. 2012, Wan et al. 2014, and Lu et al. 2022), our results do not show an inverted U-shaped relationship between variety and customer choice for Segments 1 and 2, but rather positive and negative linear relationships. Interestingly, however, we see from Table 4 that for the model using the aggregate variety measure (i.e., total products), which corresponds to the measure of variety used in the work cited above, we do see inverted U-shaped relationships for all segments. The latter observations have two important implications. First, we provide empirical support for previous studies, when using the same measure of variety. Second, however, we show that deaggregating variety into its multidimensional components significantly changes the empirical behavioural insights we obtain in this study, and further supports the idea that variety is a multidimensional construct as reported by Broniarczyk et al. (1998).

Segment 1: the *goal-directed* customers. We label Segment 1 as *goal-directed* customers (Janiszewski 1998) since they exhibit a more structured and focus session behaviour. From Table 3, we see these customers have the largest consideration sets across segments and the highest average median discount. Similarly, Moe (2003) describes these customers as “search/deliberation” customers and show the latter interact with the most number of products. Their utility for economic variables is between those of Segments 2 and 3, and our results suggest goal-directed customers are not bothered by broken assortments. While their propensity to convert is largely smaller than that of Segment 2, our results show that the utility for the “no-choice” alternative is relatively low, suggesting goal-directed customers could be converted if they find their preferred product. Although first-stage results show Segment 1 customers prefer less variety than Segment 2, summary statistics of Table 3 show they still interact with a considerable level of variety that is close to that of Segment 2. In the choice stage, goal-directed customers value more styles, less colours, and sparse assortments. Since these customers seem to prefer styles to colours, the sparse assortments they are most interested in are the ones with many styles vertices connected to only a few colours each, and hopefully the colour black, as discussed in the previous subsection. Finally, the variety constant (*delta.const*) slightly below 0 for this segment suggests a negative baseline utility for variety.

Segment 2: the *bargain-hunters*. We label Segment 2 customers as *bargain-hunters*, a special type of goal-directed customers hyper-focused on discounts. Bargain-hunters have the smallest consideration sets, which suggests a certain focus on a particular product or type of product, in addition to seeing the most products with a broken assortment ($med.cs.broken.assort = 0.911$). Recall that $broken.assort = 1$ implies that at least one size for the product is stocked-out. They are also the only segment to have a negative coefficient estimate for this variable, which suggests there could be some frustration associated to considering mostly highly discounted products as many may have the size of the customer stocked-out, and supports the idea that this segment is highly goal-oriented and has a high propensity to converting. Moe (2003) describes these “directed buying” customers and show they interact with little products, exhibit high repeat product views, and have the largest conversion rate. Bargain-hunters prefer more styles, more colours, and denser assortments in the first stage than goal-directed customers, and seem to prefer less styles, more colours and normalized density in the second stage, although Panels (a), (b) and (c) of Figure 4 suggest they are close to indifferent to all. In addition, the variety constant ($delta.const$) slightly above 0 for this segment suggests a positive baseline utility for variety.

For this segment, the bulk of the decision is made in the consider stage, where both economic (*price* and *discount*) and variety variables are most salient. The choice stage is dominated by *view.count* variable and consists of looking again and again at the products to make the final decision, and our results suggest that economic and variety variables do not play as central of a role in the choice decision as for the consideration stage. This further explains why bargain-hunters have the lowest utility for *discount* across all segments in the choice stage: within the consideration set, all products have a high level of discount, such that this variable is no longer as important in the choice stage as it is in the consider stage. The behaviour exhibited by bargain-hunters suggests they greatly benefit from larger assortments (as the latter most likely suffer from heavy-tailed distributions in which many low-selling products will get discounted) in their consideration stage, supporting with empirical evidence the findings of Kahn and Lehmann (1991). They may suffer, however, from those larger assortments when time comes to make a final decision, especially when it relates to finding the right colour for their product, as shown by the negative coefficient estimate, and discussion on the black colour presented previously.

Segment 3: the *exploratory* customers. We label Segment 3 customers as *exploratory* (Janiszewski 1998) because of their small consideration sets, lower sensitivity to economic variables (due to their session focus being more on exploring and learning the assortment rather than purchasing a product), and extremely low conversion rate. Moe (2003) describes these customers as “knowledge-building” customers and show they interact with the least products. While exploratory customers are similar to goal-directed customers (based on first-stage results), they exhibit a different behaviour in the choice stage and have the highest utility for the *price*, and *discount*, and variety variables (across segments). Furthermore, exploratory customers have the largest (across segments) utility for the “no-choice” alternative, supporting our typology that these

customers are not motivated by a purchase. Exploratory customers are impacted nonlinearly and negatively by increasing relative styles, nonlinearly and positively by relative colours, and indifferent to normalized density in that stage. Furthermore, the variety constant (*delta.const*) is lowest for this segment, suggesting a highly negative baseline utility for variety.

We find *discount* to be an important mediating variable to explain the decreasing relationship for relative styles shown in Panel (a) of Figure 4. We first test that the observed decreasing relationship is not explained by more sales (in units) coming from subcategories with lower relative styles and disprove this hypothesis by seeing more sales at medium levels (between 0.15 and 0.664) of relative styles than at low levels (between 0 and 0.15), and that this difference is significant (mean low = 0.011, mean medium = 0.058, p-value = 4.947E-07). We further test if this decreasing relationship could come from the context in which the choices are made: when a product from a subcategory with lower relative styles is in the consideration set, it is chosen over a product from a subcategory with larger relative styles. We also disprove this hypothesis by seeing that, when a product from a subcategory with a medium level of relative styles is chosen, there is *always* at least one product from a subcategory with low relative styles. This leads us to finally posit that it may be that customers in Segment 3 do value less relative styles, but that a good discount will make them pick any product over a product from a subcategory with low relative styles. We confirm this hypothesis by seeing that most of the products from subcategories with low relative styles that are chosen have little to no discount, while products coming from subcategories with medium relative styles are more heavily discounted, and that this difference is significant (mean low = 14%, mean medium = 35%, p-value = 2.268E-05). It could be that exploratory customers like smaller relative styles because it allows them to see the full variety of the subcategory, versus a subcategory with so many style levels that it is more likely than not that a customer will not see, and therefore not consider, the full offered variety within large subcategories. This evidence contrasts that of Kahn and Lehmann (1991) who say that large assortment provide more opportunities for exploratory customers to learn about the offering and their own preferences. It could be, however, that the level of variety our partner retailer operates in—as opposed to studies from a few decades ago that considered more than 10 products constitutes a large assortment—is so high that it may be beyond useful if customers are so overwhelmed with alternatives that they have trouble processing it all. On the other hand, the relationship to colours provides us with valuable insights, as it tells us that ~20 levels seem ideal for this customer segment. Once again, we find evidence to support that many customers benefit from higher levels of variety because it seems, at least for our partner retailer, that high variety leads to heavy-tailed demand and non-depleting inventory, which then leads to discounts.

Overall, our results suggest that more variety is certainly not always better. From the customer's perspective, it seems that only one segment has a true preference for variety: bargain-hunters. However, we see from Figures 3 and 4 that this preference is mostly in the consider phase, and that this segment may not value variety because of their variety-seeking predisposition (Simonson 1990) or to learn about the offered

assortment (Kahn and Lehmann 1991), but rather because it provides them with many opportunities to find discounted products. From the retailer’s perspective, then, this observation has significant implications. A retailer faces high managerial costs due to variety (high design, production, and shipping costs), but more importantly, variety makes it challenging to efficiently manage such a large inventory of products with heavy-tailed demand. Consequently, this leads to cannibalization due to discounts—since, most likely than not, highly similar products in the assortment means a customer can always find a product that is similar enough, and discounted—and overall smaller margins because, as evidenced by our results, the segment that is most likely to purchase a product also purchases highly discounted products. We also see that for the least likely segment to convert, discounts plays a central role in the final choice. We could argue then, that smaller assortments which retain high perceived variety (as suggested by Boatwright and Nunes 2001), could lead to better management, potentially few discounts, and more depleted inventories at the end of a selling season. This would contribute to reducing the inefficient use of resources caused by overproduction, while simultaneously reducing left-over inventory waste.

5 Conclusion, Limitations, and Future Work

In collaboration with one of the largest fast fashion retailers in Europe, our study aimed at understanding how assortment variety affects customer choice in large fashion assortments. We represent assortment variety as a bipartite graph, and define three measures of variety based on this representation: number of style vertices, number of colour vertices, and normalized density. We then define a customer behaviour model, formalized as a two-stage *Random Consideration Set* model directly incorporating customer heterogeneity in the modelling approach, parametrized as a Latent-Class Multinomial Logit (LC-MNL) model, and test it empirically on a large customer-level clickstream dataset. We find our dataset has three main segments, with contrasting utility for economic variables, variety, and propensity to convert to a purchase. Our estimation results also show a combination of linear and nonlinear relationship between variety and customer utility.

These findings provide empirical evidence into understanding customer behaviour, and bridging the research gaps between operations management, marketing and consumer psychology literatures. Our study further shows the importance of accounting for heterogeneity in the modelling approach, when results are compared to results of an aggregate MNL mode, and the value of deaggregating variety into its multidimensional components in the modelling approach, when results are compared to results of a LC-MNL model using an aggregate measure of variety. In the latter comparison, while model fit is comparable, we see differences in the segmentation and estimated effects of variety on customer utility.

Our segmentation shows different segments prefer different levels of variety, at different stages of their decision process, and that their preferences are misaligned (i.e. when one wants more, the other wants less). Given that different segments represent different proportions of the customer population and conversion rates, heterogeneity in customer segments suggests a careful evaluation of an assortment’s offering by the

retailer. In addition, variety is one of the many aspects that drive customer decisions. Our results show that while variety is important and significantly affects customer behaviour, it is the combination of variety and other factors (that may depend on variety, such as discounts) that drive customer decisions at our partner fast fashion retailer. The complex interactions between economic and variety variables may even benefit a certain segment for which “getting a good deal” is more important than high product fit. Managers should therefore be wary of the straightforward approach that more variety leads to more sales, as we show that in reality, more variety feeds into the heavy-tailed distribution challenge, leads to more discounted sales, and therefore additional management costs, poorer returns and an overall greater environmental footprint of their operations due to combined overproduction and wasteful left-over inventory.

A limitation to our study is that the results are only applicable to our dataset. It would be interesting to see if similar results replicate for different fast fashion retailers, or even different retail industries (e.g., grocery). Moreover, replicating this study on a dataset that has extensive product information (e.g., attributes and levels) would allow to study how different dimensions (i.e., attributes) affect customer behaviour, in addition to providing a strong empirical set up to study the effects of decluttering the attribute space as proposed by Boatwright and Nunes (2001). In addition, replicating this study on a dataset with information of sorting and filtering (i.e., a dataset with better information about product placement seen by each customer) would also provide a strong empirical set up to understand the moderating effects of placement on the perception and impact of variety on customer behaviour. This study only looked at the effects of variety on the first appearance of customers in our data. It would be interesting to see the evolution of the effects of variety on returning customers. In particular, it would be interesting to see how variety may impact deferral of purchase, or if there are any shifts as customers become more loyal to a brand or retailer, or when they transition from exploratory in one session to goal-directed in their next session.

Finally, incorporating the insights from this empirical study in a classical convex optimization formulation of the assortment optimization problem may be challenging. However, using our bipartite representation, the assortment optimization could be modeled as a network optimization problem, in a way that borrows the concept of process flexibility as proposed by Graves and Tomlin (2003). This is an interesting area of future research. Our bipartite representation can further be useful to define the state space to explore if the assortment optimization problem is solved using dynamic programming.

Appendix. Model Comparison Table

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Table 4 Results Comparison–Alternative Model Specifications

		Segment 1		Our Model Segment 2		Segment 3		Segment 1		Aggregate Variety Variable Model Segment 2		Segment 3		Aggregate MNL Model	
		Estimate	St.Err.	Estimate	St.Err.	Estimate	St.Err.	Estimate	St.Err.	Estimate	St.Err.	Estimate	St.Err.	Estimate	St.Err.
First Stage (consideration)	<i>med.cs.subcat.rel.styles</i>			0.814***	0.185	0.146	0.175								
	<i>med.cs.subcat.rel.styles²</i>			-0.394**	0.161	-0.192	0.156								
	<i>med.cs.subcat.rel.colours</i>			-1.867***	0.176	-0.117	0.174								
	<i>med.cs.subcat.rel.colours²</i>			2.571***	0.172	0.111	0.175								
	<i>med.cs.subcat.rel.norm.dens</i>			-1.888***	0.368	0.195	0.347								
	<i>med.cs.subcat.rel.norm.dens²</i>			3.424***	0.571	-0.651	0.609								
	<i>med.cs.subcat.rel.products</i>			-	-	-	-			0.294**	0.140	0.054	0.136	-	-
	<i>med.cs.subcat.rel.products²</i>			-	-	-	-			0.658***	0.150	-0.073	0.151	-	-
Second Stage (choice)	<i>rel.styles</i>	5.603**	2.422	0.950*	0.522	-61.162***	6.227	-	-	-	-	-	-	0.573***	0.068
	<i>rel.styles²</i>	0.398	2.125	-0.175	0.418	46.027***	4.953	-	-	-	-	-	-	-0.080	0.155
	<i>rel.colours</i>	1.841	3.280	-1.152**	0.518	93.386***	8.511	-	-	-	-	-	-	-0.449***	0.044
	<i>rel.colours²</i>	-11.335***	4.360	0.381	0.416	-57.834***	5.406	-	-	-	-	-	-	0.269**	0.154
	<i>rel.norm.dens</i>	16.488***	4.509	-1.829**	0.816	8.481	8.317	-	-	-	-	-	-	-1.478***	0.482
	<i>rel.norm.dens²</i>	-37.838**	16.145	1.513	1.335	4.488	48.503	-	-	-	-	-	-	1.288	1.565
	<i>med.cs.subcat.rel.products</i>	-	-	-	-	-	-	-0.291	1.661	-0.127	0.303	9.261***	1.977	-	-
	<i>med.cs.subcat.rel.products²</i>	-	-	-	-	-	-	-4.101*	2.432	0.485***	0.295	-7.449***	1.785	-	-
	<i>no.choice</i>	8.766***	0.475	5.764***	0.085	14.620***	0.951	7.571***	0.384	5.924***	0.075	18.948***	1.336	6.258***	0.006
	Segment size	5.55%		23.50%		70.95%		9.79%		25.79%		64.42%		-	
	Conversion rate	0.48%		8.23%		0.085%		0.26%		7.57%		0.065%		-	
Log-likelihood				-20,345.39						-20,401.42				-21,152.28	
BIC				20,717.62						20,653.58				21,236.34	
McFadden R ²				0.943						0.943				0.941	

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Electronic Companion to “Understanding the multidimensional effects of product variety on customer behaviour: A large-scale empirical study in online fast fashion retail”

EC.1 Data Preparation, Variable Definition & Sample Data

EC.1.1 Data Preparation

Recall that we are only interested in *dress shoppers*, i.e. customers that have at least one dress product in their consideration set. Additionally, our data collection and cleaning serves two purposes: obtaining structured data of the full session of customers for clustering purposes, and obtaining structured data relative to the dress category only for estimation purposes. We extract from the raw clickstream data, for each user, the list of product view, basket operation, and purchase events only for dress shoppers. If a customer leaves the online store without making a purchase, we assign the no-purchase option as the purchased product. In the event a dress shopper makes a purchase, but in a different category than the dress category (e.g., pants), we modify the sales event to represent the no-purchase (in the dress category) option.

We filter the customers in our data to retain only those that have interacted with five dress products or more, allowing us to collect customers with a sufficient number of interactions for the clustering and estimation purposes. This decision is made as we assumed sessions for customers interacting with less than five products do not possess enough information to provide insights into customer choice. Moe (2003) empirically shows that customers interacting with on average 2 pages were clustered together, and did not provide insights in terms of customer behaviour or typology. We choose a cutoff at 5 products as we feel, based on preliminary inspection of our dataset, that a cutoff at 10 products would be too limiting, and two products would still leave us with too much noise in the data. Applying this cutoff rule results in removing 66% of the unique customers in the dataset.

Furthermore, as mentioned previous, we are only interested in applying our model to each customers' first visit to the online store, and consequently remove any subsequent visits from returning customers other than the first one.

We also modify our dataset to retain customers that purchased more than one dress product from a same sales event (e.g., a customer purchases 3 different dresses on her first visit). Since the parametrization we use in our study assumes customers can only choose one product at a time (i.e. purchase exactly one unit of one product), in those few cases ($\sim 1\%$ of users), we create as many copies of the consideration set as unique products purchased, and assume separate purchase events for each. Aouad et al. (2019) use the same approach. For the few cases ($< 1\%$ of users) in which customers purchase more than one unit of a same product, we modify the sales event to be of one unit.

To have the information contained in the firm-side data at the same level as the clickstream data, the first step is to aggregate, for each day, the SKU information at the product level, i.e. aggregating information of all sizes of a product.

EC.1.1.1 Firm Data Imputation.

We noted that a large fraction of the products ($\sim 65\%$) had missing entries for one or more dates of their selling season. We assume that the missing entries corresponds to dates for which there were no transactions (either customer purchases or inventory restock), and impute the data with those missing entries based on the last available date. This allows us to have continuous entries for each product in our dataset, which is necessary for the computation of several variables and estimation of the model.

For example, for a hypothetical product that is put on the online store on 2018-02-20, we see daily entries the first week of availability of the product (from 2018-02-20 to 2018-02-27), 5 subsequent dates missing (2018-02-28 to 2018-03-04), and then daily entries for the remainder of the time it is available (from 2018-03-05 to 2018-06-24). We impute the data by copying, for each missing date, the row corresponding to the entry of 2018-02-27.

EC.1.2 Outliers

Upon preliminary inspection of our dataset, we noted the extreme long-tail nature of the distribution of the size of the consideration set of each customer (the largest consideration set has 8,176 products). Because our data is collected from our partner retailer's website, we were concerned some data might not necessarily have been generated by humans, but rather by bots or web crawlers. Note here that, as opposed to other applications (e.g., clinical trials), what we call outliers here are neither measurement errors nor novel data. They are data points generated outside the data generating process we aim to study, i.e. human customers shopping online. Therefore, we remove those data points.

To identify outliers, we follow the method proposed by Hubert and Vandervieren (2008) which applies specifically to skewed distributions. Applying this rule, we remove any customer that has more than 72 products in her (full) consideration set (1% of the unique customers), or more than 22 products in her dress consideration set (3% of the unique customers).

EC.1.3 Sample Data and Summary Statistics

Tables EC.1 presents sample first-stage data, EC.2 presents summary statistics of first-stage variables, and EC.3 presents the correlation matrix of first-stage variables.

Tables EC.4 presents sample second-stage data, EC.5 presents summary statistics of second-stage variables, and EC.6 presents the correlation matrix of second-stage variables.

Table EC.1 Sample First-Stage Estimation Data

<i>userId</i>	<i>date</i>	<i>consideration.set</i>	<i>med.cs.price</i>	<i>med.cs.discount</i>	<i>med.cs.age</i>	<i>med.cs.view.count</i>	<i>med.cs.broken.assort</i>	<i>med.cs.rel.styles</i>	<i>med.cs.rel.colours</i>	<i>med.cs.norm.dens</i>
user hash 1	2018-02-19	5	79.99	0	17	1	0	0.277	0.308	0.06
user hash 2	2018-02-19	16	99.99	0.434	76	1	1	0.277	0.308	0.05
user hash 3	2018-02-19	11	169.99	0.576	76	2	1	0.253	0.194	0.021
user hash 4	2018-02-19	10	129.99	0.249	20.5	7	1	0.159	0.194	0
user hash 5	2018-02-19	7	149.99	0.573	76	1	0	0.253	0.156	0.021

Table EC.2 First-Stage Variables–Summary Statistics

	Mean	St. Dev.	Min.	Max.
<i>med.cs.price</i>	127.47	39.67	24.99	299.99
<i>med.cs.discount</i>	0.23	0.20	0	0.71
<i>med.cs.age</i>	70.074	40.966	1	173
<i>med.cs.rel.styles</i>	0.297	0.163	0	1
<i>med.cs.rel.colours</i>	0.383	0.170	0	1
<i>med.cs.rel.norm.dens</i>	0.027	0.035	0	1

Table EC.3 First-Stage Variables–Correlation Matrix

	<i>med.cs.price</i>	<i>med.cs.discount</i>	<i>med.cs.age</i>	<i>med.cs.rel.styles</i>	<i>med.cs.rel.colours</i>	<i>med.cs.rel.norm.dens</i>
<i>med.cs.price</i>	1	-0.060	-0.155	0.045	-0.214	-0.252
<i>med.cs.discount</i>	-	1	0.707	-0.041	-0.111	-0.043
<i>med.cs.age</i>	-	-	1	0.012	0.011	-0.033
<i>med.cs.rel.styles</i>	-	-	-	1	0.686	-0.350
<i>med.cs.rel.colours</i>	-	-	-	-	1	-0.246
<i>med.cs.rel.norm.dens</i>	-	-	-	-	-	1

Table EC.4 Sample Second-Stage Estimation Data

<i>userId</i>	<i>date</i>	<i>productId</i>	<i>choice</i>	<i>price</i>	<i>discount</i>	<i>age</i>	<i>view.count</i>	<i>broken.assort</i>	<i>rel.styles</i>	<i>rel.colours</i>	<i>rel.norm.dens</i>
user 1 hash	2018-02-19	product 1 hash	0	149.99	0.47	129	1	1	0.098	0.203	0.049
user 1 hash	2018-02-19	product 2 hash	0	179.99	0.00	63	1	1	0.358	0.543	0.021
user 1 hash	2018-02-19	product 3 hash	0	149.99	0.57	129	1	1	0.188	0.271	0.027
user 1 hash	2018-02-19	product 4 hash	0	249.99	0.48	129	1	0	0.188	0.271	0
user 1 hash	2018-02-19	product 5 hash	1	199.99	0.50	129	1	1	0.358	0.543	0.457

Table EC.5 Second-Stage Variables–Summary Statistics

	Mean	Std. Dev.	Min.	Max.
<i>price</i>	118.01	64.59	1.00	299.990
<i>discount</i>	0.33	0.32	0	1
<i>age</i>	61.996	49.179	0	173
<i>view.count</i>	1.198	0.607	1	36.000
<i>num.styles</i>	31.897	28.443	1	115
<i>tot.styles.dresses</i>	671.714	235.360	1	835
<i>rel.styles</i>	0.309	0.273	0	1
<i>num.colours</i>	11.695	7.685	1	29
<i>tot.colours.dresses</i>	58.905	19.963	1	67
<i>rel.colours</i>	0.405	0.265	0	1
<i>norm.dens</i>	0.019	0.041	0	0.5
<i>med.norm.dens.dresses</i>	0	0.0040	0	0.015
<i>rel.norm.dens</i>	0.031	0.071	0	1

A note on colour grouping (for *tot.colours* and *norm.density*). The retailer has a total of 3,478 different colour codes for the seasons we restrict our application to. Those are highly specific colour codes that have a product-differentiation purpose on the retailer’s end more than for customers. On the online store, products of the dress category are categorized into 80 different colours that are found in the product descriptions. Those colours cover the full range of “basic” colours (e.g., red, green, blue, black, white, etc.), but also

Table EC.6 Second-Stage Variables–Correlation Matrix

	<i>choice</i>	<i>price</i>	<i>discount</i>	<i>age</i>	<i>view.count</i>	<i>broken.assort</i>	<i>rel.styles</i>	<i>rel.colours</i>	<i>rel.norm.dens</i>
<i>choice</i>	1	-0.601	0.705	-0.416	-0.092	0.202	-0.373	-0.504	-0.146
<i>price</i>	-	1	-0.410	0.240	0.055	-0.216	0.201	0.212	-0.053
<i>discount</i>	-	-	1	0.088	-0.093	0.337	-0.272	-0.407	-0.184
<i>age</i>	-	-	-	1	0.042	0.251	0.198	0.235	-0.002
<i>view.count</i>	-	-	-	-	1	-0.019	0.025	0.062	0.025
<i>broken.assort</i>	-	-	-	-	-	1	-0.089	-0.150	0.019
<i>rel.styles</i>	-	-	-	-	-	-	1	0.826	-0.237
<i>rel.colours</i>	-	-	-	-	-	-	-	1	-0.176
<i>rel.norm.dens</i>	-	-	-	-	-	-	-	-	1

include specific colours (e.g., mint, navy, bordeaux, etc.) as well as patterns (e.g., black striped, black patterned, turquoise check, etc.). Finally, the retailer also has a higher-level categorization of colours with 15 distinct categories. Those include only “basic” colours.

To define the subcategory variety variables *tot.colours* and *norm.density*, we used the middle level of 80 colours. This allows to retain enough flexibility in the definition of the different colours a customer can perceive (black, black striped and black patterned are assumed to be all distinct colour/patterns). From our data, we see that any subcategory never has over 30 of the 80 colours included.

EC.2 EM Algorithm

Algorithm 1 presents the pseudo-code of the EM algorithm implemented to estimate our model coefficients.

We use a Newton-Raphson (NR) algorithm with inexact linesearch to solve each substep. We choose this second-order method over gradient-based methods (e.g., gradient descent, BHHH) because of the large scale differences in model variables, which caused convergence issues. We decided against normalizing variables to preserve interpretability of the results. Since our parameter space is relatively small, the NR algorithm converges rapidly (less than 5 iterations, taking only a few minutes).

EC.3 Endogeneity Results

Tables EC.7 presents the regression results of the first-stage regression. Note that the first stage regression is at the product level, and does not include any customer-specific data (*view.count* variable).

The first stage results show a high goodness-of-fit ($R^2 = 0.941$ for *discount* and $R^2 = 0.864$ for *broken.assort*), and therefore validate the instruments for their use in the control function approach. *discount.lag* explains 94% of the variance (i.e. all the variance) in the *discount* model (Column (1)), while *broken.assort.lag* explains 86% of the variance (i.e., all the variance) in the *broken.assort* model (Column (2)), showing that the chosen instruments significantly contribute to the prediction of the endogenous variables.

EC.4 Additional Estimation Results

EC.4.1 MNL Model

Table EC.8 presents the estimation results for an aggregate MNL model using our proposed variety variable.

Algorithm 1 EM Algorithm for LC-MNL Estimation

```

1: Initiate  $e = 0$ ,  $\kappa^0 = \mathbf{0}$ , that is, assign an equal segment membership probability to all customers.
2: Partition data into  $M$  groups. Solve a separate MNL for each group to obtain initial values for  $\beta^{m,0}, \delta^{m,0}, \eta^{m,0}$ . The partitioning is independent of the latent segmentation in the data, and is only made to obtain different values for each segment to initiate the algorithm (Train 2009).
3: while  $\Delta \log - \text{likelihood} \geq \varepsilon$  do
4:   STEP 1:
5:   for segment  $m$  do
6:     for customer  $i$  do
7:       Compute the posterior probability of customer  $i$  to be in segment  $m$  given observed choices as:

$$H_{im}^{e+1} = \frac{h_{it}^e(m) Q_{it}^{m,e}}{\sum_m h_{it}^e(m) Q_{it}^{m,e}}$$

       where  $Q_{it}^{m,e} = \prod_{j \in C_{it}} q_{ij}(j|m)^{y_{ijt}}$  corresponds to the choice likelihood of customer  $i$  given segment  $m$  membership of the previous iteration.
8:     end for
9:   end for
10:  STEP 2:
11:  Update coefficients  $\kappa_m^{e+1}$  as:

$$\kappa_m^{e+1} = \arg \max_{\kappa} \prod_t \prod_i \prod_m h_{it}(m)^{H_{im}^{e+1}}$$

       where the dependent variable of this MLE of the MNL model for segment membership is the vector of posterior segment membership probabilities  $H_{im}^{e+1}$ .
12:  Update segment membership probabilities  $h_{it}^{e+1}(m)$  as:

$$h_{it}^{e+1}(m) = \frac{\exp\{z_{it} \kappa^{m,e+1}\}}{\sum_{l=1}^M \exp\{z_{it} \kappa^{l,e+1}\}}$$

13:  STEP 3:
14:  for segment  $m$  do
15:    Update the coefficients  $\beta^{m,e+1}, \delta^{m,e+1}, \eta^{m,e+1}$  by solving a separate MNL model for segment  $m$ . Each MNL model uses the full data and weights each respondent by  $H_{im}^{e+1}$ . Furthermore, each segment-specific MNL model includes the control function approach to correct for potential endogeneity in the discount and broken assort variables (see §3.3.2).
16:    Update likelihood  $Q_{it}^{m,e+1}$  given  $\beta^{m,e+1}, \delta^{m,e+1}, \eta^{m,e+1}$ .
17:  end for
18: end while

```

EC.4.2 LC-MNL Model—Aggregate Variety Measure (*rel.prods*)

Table EC.9 compares the BIC for each model. Table EC.10 presents the estimation results for a LC-MNL model specification in which the only variety variable included is the total number of product (i.e., the aggregated measure of variety).

Table EC.7 First-Stage Regression Estimation Results

	<i>Dependent variable:</i>	
	<i>discount</i> (1)	<i>broken assort</i> (2)
<i>price</i>	2.62E-03 (5.287E-04)	-5.09E-05 (1.661E-05)
<i>age</i>	1.57E-03 (0.001)	1.17E-04 (1.712E-05)
<i>broken assort</i>	-0.132 (0.128)	-
<i>discount</i>	-	-1.30E-04 (1.259E-04)
<i>rel.styles</i>	-0.381 (0.240)	1.51E-03 (7.552E-03)
<i>rel.styles</i> ²	1.197 (0.450)	0.012 (0.014)
<i>rel.colours</i>	-1.31E-03 (0.196)	-0.011 (6.165E-03)
<i>rel.colours</i> ²	-0.190 (0.443)	7.94E-04 (0.014)
<i>rel.norm.dens</i>	-1.844 (0.724)	-4.04E-03 (0.023)
<i>rel.norm.dens</i> ²	0.091 (1.200)	0.023 (0.038)
<i>resid.discount</i>	0.964 (1.169E-03)	3.44E-04 (1.267E-04)
<i>resid.ba</i>	-0.017 (0.127)	0.918 (1.566E-03)
Observations	62,410	62,410
R ²	0.940	0.865
Adjusted R ²	0.939	0.865
F Statistic (df = 11; 62,301)	87,989.060***	36,047.150***

*: $p \leq 0.1$; **: $p \leq 0.05$; ***: $p \leq 0.01$; (std dev)**Table EC.8** Estimation Results–MNL Model

	Estimate	Std.Err.
<i>price</i>	-7.909E-03***	2.17E-07
<i>discount</i>	0.012***	1.00E-06
<i>age</i>	8.762E-04***	2.30E-07
<i>broken assort</i>	-0.304***	2.05E-03
<i>view.count</i>	0.632***	7.26E-05
<i>rel.styles</i>	0.573***	0.068
<i>rel.styles</i> ²	-0.080	0.155
<i>rel.colours</i>	-0.449***	0.044
<i>rel.colours</i> ²	0.269*	0.154
<i>rel.norm.dens</i>	-1.478***	0.482
<i>rel.norm.dens</i> ²	1.288	1.565
<i>resid.discount</i>	-2.579E-03***	1.52E-05
<i>resid.ba</i>	0.239***	9.47E-03
<i>no.choice</i>	6.258***	6.08E-03
Observations	1,595,492	
Log-likelihood	-21,152.28	
BIC	21,236.34	
McFadden R ²	0.941	

*: $p \leq 0.1$; **: $p \leq 0.05$; ***: $p \leq 0.01$

EC.5 Additional Tables and Figures

Table EC.11 below presents the distribution of units sold in the dress category for the 3-month period of interest for this study by colour groups.

Table EC.9 Model Fit Comparison–Aggregate Variety Measure (*rel.prods*)

M	2	3	4
BIC	20,711.83	20,653.58	20,668.96

Table EC.10 Estimation Results–Aggregate Variety Measure (*rel.prods*)

Stage	Variable	Segment 1		Segment 2		Segment 3	
		Estimate	St.Err.	Estimate	St. Err.	Estimate	St. Err.
First Stage (consideration)	<i>med.cs.price</i>			-3.747E-03***	1.672E-04	1.662E-04	1.514E-04
	<i>med.cs.discount</i>			0.016***	4.357E-04	-1.475E-03***	4.224E-04
	<i>med.cs.age</i>			-1.170E-03***	2.159E-04	3.951E-04*	2.098E-04
	<i>med.cs.subcat.rel.products</i>			0.294**	0.140	0.054	1.362E-01
	<i>med.cs.subcat.rel.products</i> ²			0.658***	0.150	-0.073	1.505E-01
	constant		Base segment	-0.153***	0.035	-8.247E-03	3.223E-02
Second Stage (choice)	<i>price</i>	-0.023***	2.833E-03	-4.650E-03***	4.311E-04	-0.028***	4.044E-03
	<i>discount</i>	5.190E-03	5.106E-03	9.680E-03***	1.052E-03	-3.530E-03	8.480E-03
	<i>age</i>	0.012***	2.252E-03	-8.755E-04*	5.094E-04	0.105***	8.161E-03
	<i>borken assort</i>	-0.340	0.252	-0.298***	0.048	-0.885***	0.278
	<i>view.count</i>	0.414***	0.036	1.003***	0.012	0.279***	0.049
	<i>rel.products</i>	-0.291	1.661	-0.127	0.303	9.261***	1.977
	<i>rel.products</i> ²	-4.101*	2.432	0.485*	0.295	-7.449***	1.785
	<i>delta.const</i>	0.494	0.315	-7.925E-03	0.067	-2.171***	0.450
	<i>resid.discount</i>	0.062***	8.471E-03	-8.212E-03*	4.267E-03	-0.033***	0.016
	<i>resid.ba</i>	-3.229***	0.299	0.543***	0.097	0.845	0.770
	<i>no.choice</i>	7.571***	0.384	5.924***	0.075	18.948***	1.336
Segment size		9.79%		25.79%		64.42%	
Conversion rate		0.26%		7.57%		0.065%	
Observations		1,595,492					
Log-likelihood		-20,401.42					
McFadden <i>R</i> ²		0.943					

*: $p \leq 0.1$; **: $p \leq 0.05$; ***: $p \leq 0.01$ **Table EC.11** Distribution of Units Sold by Colour

Colour	Sales (in units)	
Black	1295	41.09%
Red	239	7.58%
Navy Blue	235	7.46%
Pink	233	7.39%
Gray	200	6.35%
Green	186	5.90%
Blue	164	5.20%
White	144	4.57%
Yellow	143	4.54%
Burgundy	79	2.51%
Coffee	73	2.32%
Ecru	67	2.13%
Mixed	35	1.11%
Purple	33	1.05%
Orange	26	0.82%

References

- Hubert M, Vandervieren E (2008) An adjusted boxplot for skewed distributions. *Computational Statistics & Data Analysis* 52(12):5186–5201,
- Train KE (2009) EM Algorithm. *Discrete Choice Methods with Simulations*, 347–370.

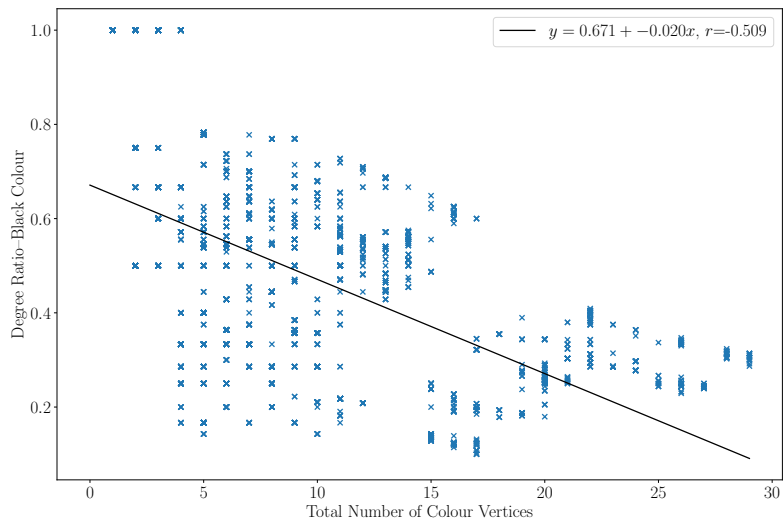


Figure EC.1 Relationship between Number of Colour Vertices and Degree of Black Colour Vertex